

Final Report

Project Title:

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Interactive Visualization Dashboard for Real Estate Sale Price and Property Feature Insights: A Case Study

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1. Introduction

1.1 Project Overview

The real estate sector generates vast volumes of transactional data covering sale prices, property age, renovation history, and structural features such as bedrooms, bathrooms, and floors. Much of this information remains locked inside spreadsheets that are difficult to interpret at a glance. This project focuses on the design and development of an interactive Tableau-based visualization dashboard that enables users such as real estate analysts, marketing professionals, and property decision-makers to explore housing sale trends and understand how structural and renovation-related features influence overall property value.

1.2 Purpose

- Visualize key housing metrics such as sale price, house age, renovation status, and structural features (bedrooms, bathrooms, floors).
- Analyze how years since renovation and house age relate to overall market behaviour and sale price.
- Assess the impact of renovations on property value to support smarter pricing decisions.
- Deliver actionable insights through interactive dashboards and storyboards that support strategic decision-making, optimize pricing models, and enhance market competitiveness.
- Serve a broad range of stakeholders, including real estate analysts, marketing professionals, and decision-makers seeking a clearer understanding of housing market dynamics.

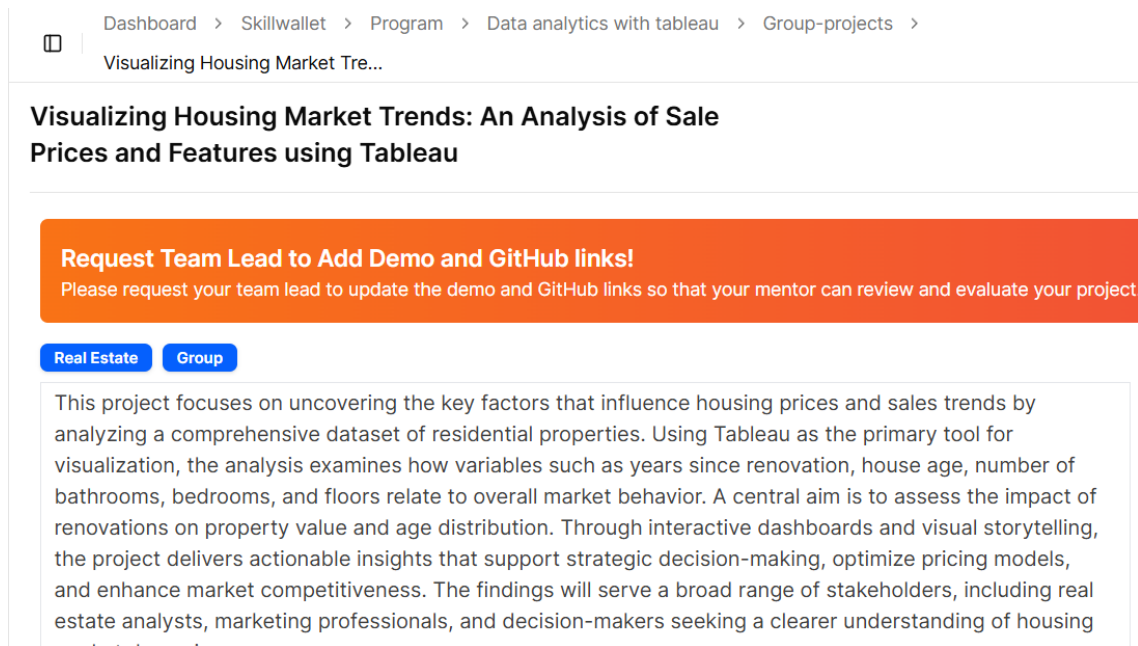


Fig 1.1 — Project overview as tracked on the SkillWallet project workspace

2. Ideation Phase

2.1 Problem Statement

Problem Statement: Understanding Housing Market Trends for Better Decision-Making

The real estate market generates large volumes of sales data, yet this data is rarely leveraged effectively due to a lack of accessible visualization and analysis tools. Stakeholders often rely on static spreadsheets or fragmented reports, making it difficult to identify pricing patterns, understand the effect of renovations on property value, or benchmark similar properties against one another.

There is a growing need for an interactive analytics solution that consolidates residential sales data — covering variables such as years since renovation, house age, number of bathrooms, bedrooms, and floors — into a single, easy-to-navigate dashboard. Such a solution helps real estate analysts, marketing teams, and decision-makers identify what drives housing prices, optimize pricing models, and respond faster to shifting market conditions.

2.2 Empathy Map Canvas

Empathy Map: Real Estate Analyst (Focused on Understanding Market Trends)

SAYS	“I want to know which features actually drive up sale price.” “I need to compare renovated vs. non-renovated homes quickly.”
THINKS	“Is an older, renovated house more valuable than a newer, unrenovated one?” “How do I make this data easy for my team to understand?”
DOES	Pulls raw sales data from the database and imports it into Tableau. Builds dashboards and filters to compare houses by age, floors, and renovation status.
FEELS	Confident when trends are clear and actionable. Frustrated when data is messy or scattered across multiple sources.
GAINS	A single source of truth for housing sales performance. Faster, more confident pricing and marketing decisions.
PAIN POINTS	Manually cross-referencing spreadsheets is time-consuming. Difficult to spot patterns without visual tools.

2.3 Brainstorming

Project Overview: As residential markets grow more data-rich, understanding which property characteristics influence sale price is key to smarter pricing and marketing. This project proposes an interactive Tableau dashboard that helps analysts and decision-makers visualize sales data, renovation trends, and structural features.

Brainstorming Process

- Step 1: Team Gathering and Collaboration — form a multidisciplinary team and select a focused problem statement.
- Step 2: Idea Listing and Grouping — generate ideas from all team members and cluster similar ideas (data collection, visualization, storytelling).
- Step 3: Idea Prioritization — rank ideas by impact and feasibility, and select the top ideas to implement in the project.

Scenario 1: Real Estate Analyst Benchmarking Renovated Homes

An analyst at a property consultancy wants to know whether recently renovated homes sell for significantly more than older, unrenovated ones. Using the dashboard, they filter by “years since renovation” and instantly see that homes renovated within the last few years command a noticeably higher average sale price.

Scenario 2: Marketing Team Planning a Listing Campaign

A marketing team preparing a campaign for a new listing uses the dashboard to check how similar houses — matched by age, floor count, and bathroom count — have performed in past sales, helping them position the new listing competitively.

3. Requirement Analysis

3.1 Solution Requirement

Functional Requirement:

#	Requirement	Details
1.	Data Collection Interface	• Import residential property sales dataset from cloud storage (Google Drive). • Connect the data source directly to Tableau for live refresh.
2.	Data Storage & Management	• Centralized cloud-based storage for raw housing sales data. • Capability to update records as new listings/sales are added.
3.	Data Cleaning & Preprocessing	• Remove duplicates and null values. • Standardize fields such as renovation year, floors, and bathroom/bedroom counts.
4.	Interactive Visualization (Tableau)	• Dashboards showing sale price trends, house age distribution, and renovation impact. • Filters by house age, renovation status, and structural features. • Trend analysis and storyboards.
5.	Analytics & Insights	• Pattern recognition (e.g., price bands most affected by renovation). • Comparison of renovated vs. non-renovated homes across age groups.

Non-Functional Requirement:

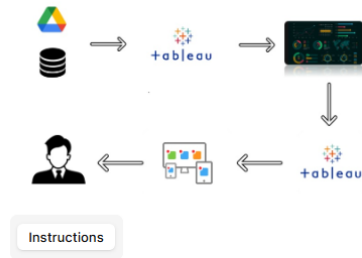
#	Requirement	Details
1.	Scalability	Should handle sales records for hundreds or thousands of properties at a time.
2.	User-Friendliness	Dashboard and reports should be easy to navigate, with minimal training required.
3.	Performance	Fast data processing and dashboard loading, even for large datasets.
4.	Data Privacy & Security	Secure login and role-based access control to sales and property data.
5.	Compatibility	Should work on various devices (laptops, tablets, mobiles) and support data export (PDF, Excel).

3.2 Data Flow Diagram

Raw housing sales data stored in Google Drive / a database is imported and modeled in Tableau, then rendered as interactive dashboards. These dashboards are viewed across desktop, tablet, and mobile devices by end users — analysts, marketers, and decision-makers — who interact with filters that feed back into Tableau for refreshed views.

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Technical Architecture



To accomplish this, we have to complete all the activities listed below,

- Data Collection
 - Collect the dataset,
 - Connect Data with Tableau
- Data Preparation
- Prepare the Data for Visualization
- Data Visualizations
 - Visualizations
- Dashboard
 - Responsive and Design of Dashboard
- Storyboard
 - Storyboard Creation
- Performance Testing
 - Utilization of Data Filters
 - No. of Calculation fields
 - No. of Visualizations/Graphs
- Web Integration

Fig 3.1 — Technical architecture and activity breakdown tracked for the module

3.3 Technology Stack

Stage	Tool/Platform	Purpose
1. Data Source	Google Drive / Database	Source of raw residential sales data (price, age, renovation status, structural features)
2. Data Cleaning & Prep	Microsoft Excel	Removed duplicates & nulls; categorized renovation status; created calculated fields and pivot tables
3. Documentation & KPIs	MS Word / Google Docs	Documented requirements, KPI definitions, and calculation logic
4. Visualization	Tableau	Created dashboards & charts; built a Tableau Story for insights; used filters and drill-down for deeper analysis
5. Story Telling	Tableau Story	Presented key findings in narrative format; used story points for a step-by-step flow

4. Project Design

4.1 Problem Solution Fit

Problem	Solution
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Lack of a consolidated view of what drives housing prices.	Build interactive Tableau dashboards consolidating price, age, renovation, and structural data.
Difficulty identifying the impact of renovation on property value.	Use “years since renovation” filters and visual comparisons to isolate the renovation effect.
Manual, spreadsheet-based analysis is slow and error-prone.	Automate data cleaning and visualization through Tableau, reducing turnaround time.
Insights are not easily accessible to non-technical stakeholders.	Use dashboards and storyboards to simplify communication of market trends.

4.2 Solution Architecture

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve the existing housing analytics problem.
- Describe the structure, characteristics, behaviour, and other aspects of the dashboard to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Technical Architecture: Data (Kaggle / Google Drive) → Data Cleaning & Prep (Excel) → Documentation & KPIs (Word / Docs) → Visualization (Tableau).

5. Project Planning & Scheduling

5.1 Project Planning

The activity breakdown for the module (see Fig 3.1) spans 8 Epics, 13 Tasks, and 6 Subtasks, covering Data Collection, Data Preparation, Data Visualization, Dashboard, Storyboard, Performance Testing, and Web Integration.

Sprint	Functional Requirement	User Story #	User Story / Task	Story Point	Priority
Sprint-1	Data Collection	USN-1	As a user, I can collect the housing sales dataset and connect it with Tableau.	2	High
Sprint-1	Data Preparation	USN-2	As a user, I can prepare the data for visualization.	1	High
Sprint-2	Data Visualization	USN-3	As a user, I can create visualizations to explore sale price and feature trends.	2	High
Sprint-2	Dashboard	USN-4	As a user, I can design a responsive dashboard.	2	High
Sprint-3	Storyboard	USN-5	As a user, I can create a storyboard to narrate key insights.	2	Medium
Sprint-3	Performance Testing	USN-6	As a user, I can test filters, calculation fields, and visualizations.	2	High
Sprint-4	Web Integration	USN-7	As a user, I can publish and embed the dashboard and story with a Flask UI.	2	Medium

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Point	Duration	Sprint Start Date	Sprint End Date	Story Point Completed	Release Date
Sprint-1	20	5 Days	20 June 2026	25 June 2026	20	25 June 2026
Sprint-2	20	5 Days	26 June 2026	01 July 2026	18	01 July 2026
Sprint-3	20	5 Days	02 July 2026	07 July 2026	17	07 July 2026
Sprint-4	20	5 Days	08 July 2026	13 July 2026	20	13 July 2026

Velocity: With a 5-day sprint duration and a team velocity of 20 story points per sprint, the average velocity (AV) works out to $AV = \text{sprint duration} \div \text{velocity} = 5 \div 20 = 0.25$ sprints per point (i.e., the team completes 4 points per day on average).

6. Functional and Performance Testing

Model Performance Testing:

S.No	Parameter	Screenshot / Value
1.	Records Rendered	21,609
2.	Data Preprocessing Steps	12
3.	Utilization of Data Filters	6
4.	No. of Calculation Fields Used	8
5.	No. of Visualizations / Graphs	4

7. Results

7.1 Output Screenshots

Total Sales by Years Since Renovation

This histogram bins total sale price by the number of years since renovation, showing that recently renovated homes (0–1 years) account for a disproportionately large share of total sales value, while sales taper off as renovation age increases.

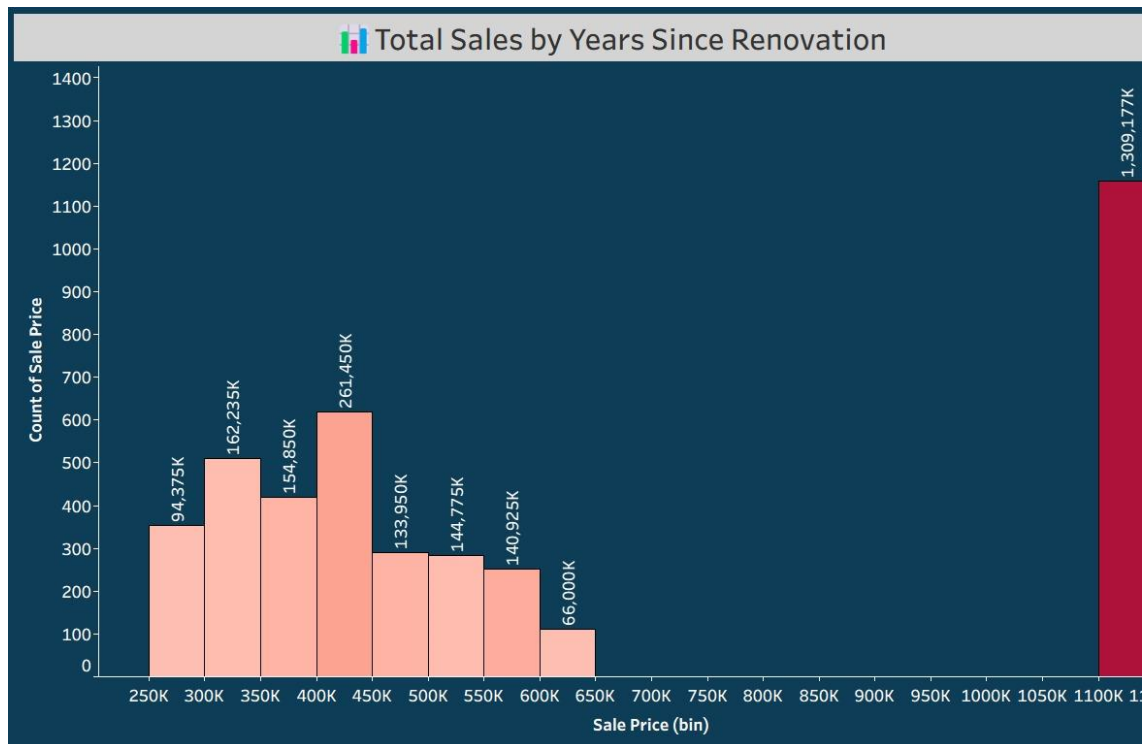


Fig 7.1 — Total Sales by Years Since Renovation

Distribution of House Age by Renovation Status

This pie chart breaks down the count of properties by house age band, illustrating how renovation activity is spread across houses of different ages.

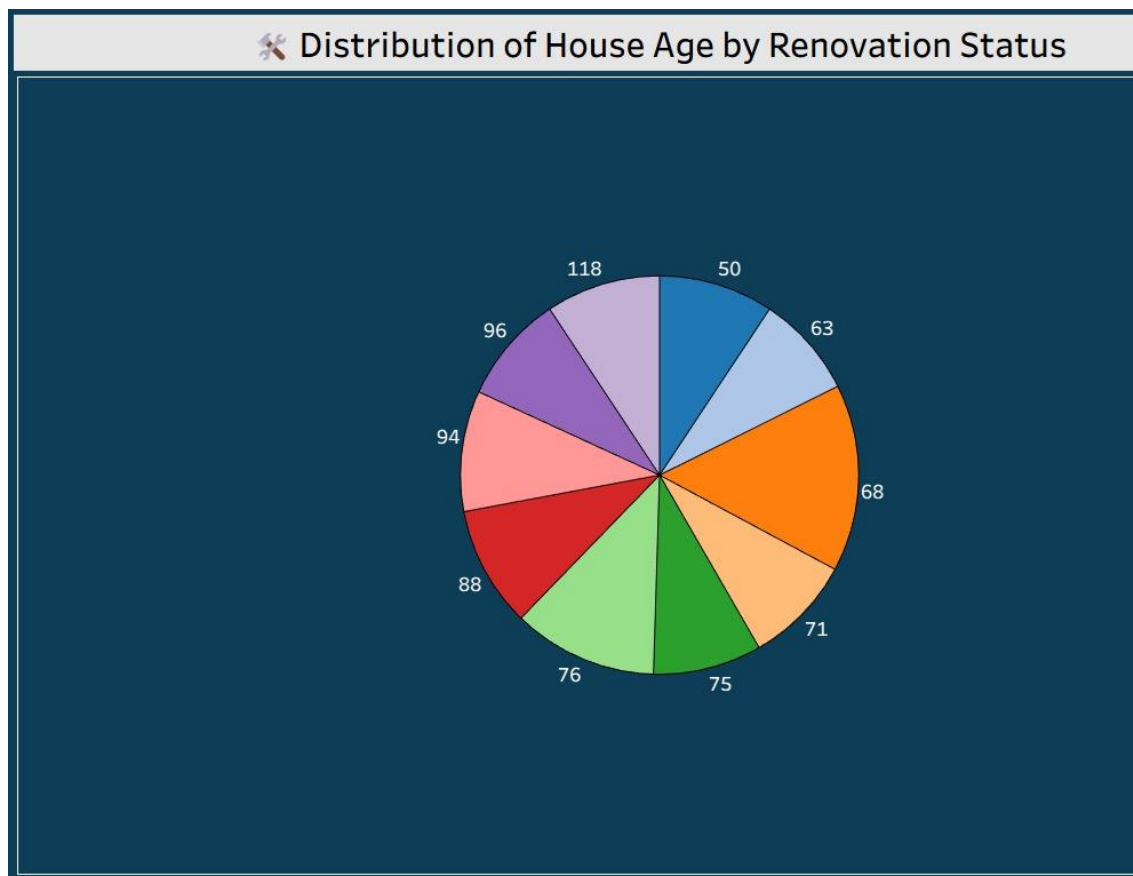


Fig 7.2 — Distribution of House Age by Renovation Status

House Age Distribution (Bathrooms, Bedrooms, Floors)

This bar chart compares the number of bathrooms, bedrooms, and floors across different house ages, helping identify how structural features evolve with the age of the property.

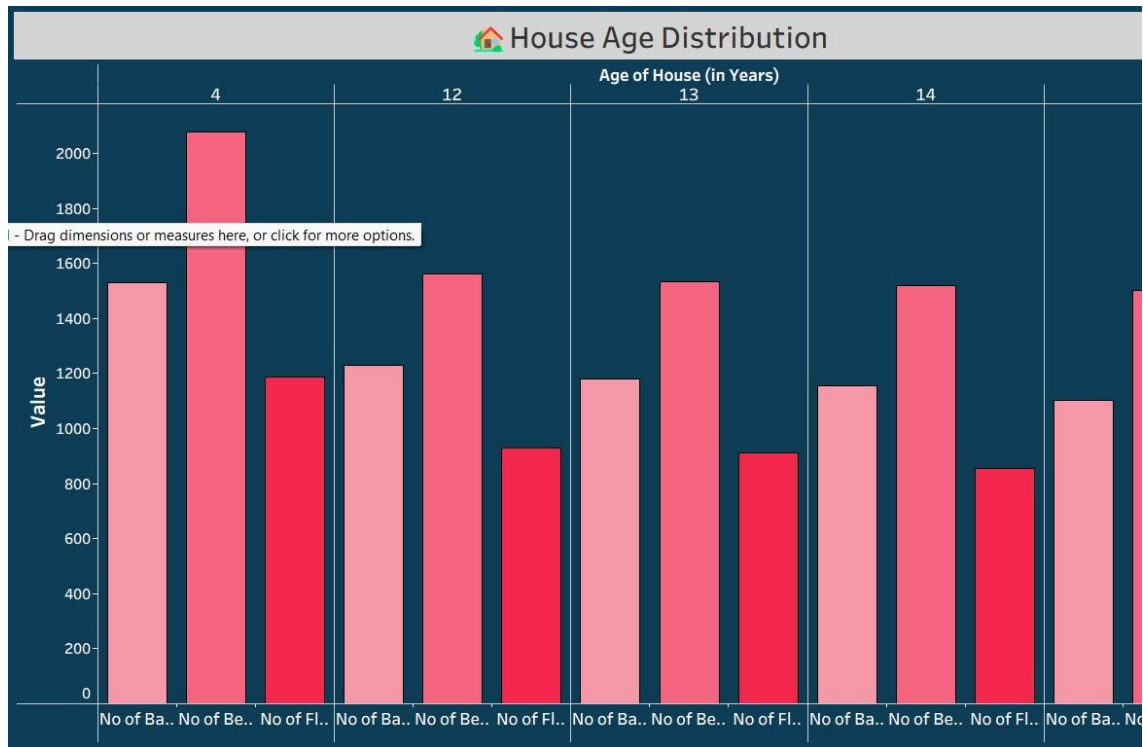


Fig 7.3 — House Age Distribution by Bathrooms, Bedrooms and Floors

Housing Sales Analysis Dashboard

The consolidated dashboard combines KPI tiles (area of house, count of transformed records, and average sale price) with the sales, renovation, and house-age visualizations into a single interactive view for end users.

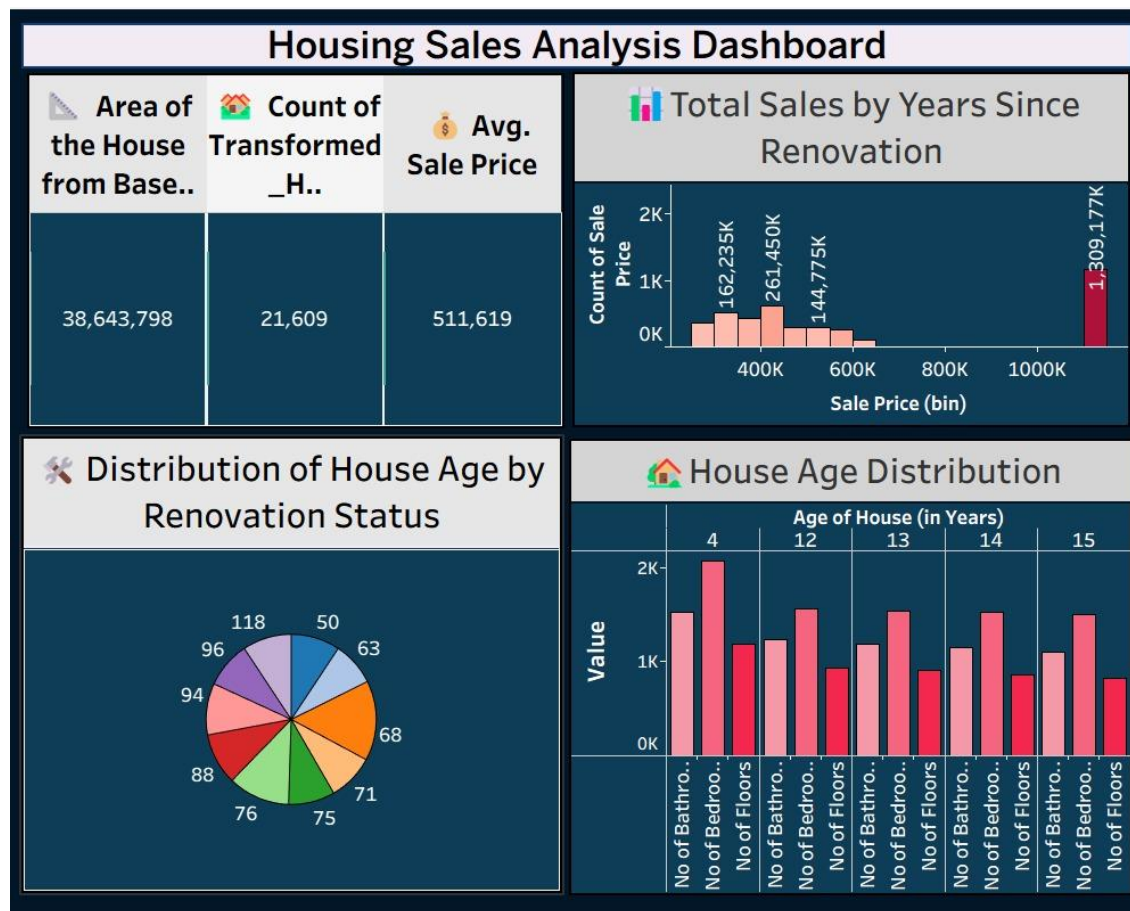


Fig 7.4 — Housing Sales Analysis Dashboard (combined view)

8. Advantages & Disadvantages

Advantages:

- **Data-Driven Decisions:** Enables pricing and marketing decisions grounded in real sales data rather than guesswork.
- **Faster Insights:** Interactive filters and dashboards replace slow, manual spreadsheet analysis.
- **Clear Renovation Impact:** Makes it easy to see how renovation history affects sale price.
- **Accessible to Non-Technical Users:** Visual storytelling makes trends understandable for stakeholders without a data background.
- **Scalable:** Can accommodate growing volumes of sales records as new data becomes available.

Disadvantages:

- **Data Quality Dependency:** Insights are only as good as the underlying dataset; missing or inconsistent records can skew results.
- **Limited Predictive Power:** Dashboards describe historical trends but do not forecast future prices without additional modeling.
- **Manual Refresh Overhead:** Without an automated pipeline, updating the dataset requires manual re-import and cleaning.
- **Learning Curve:** Building custom calculated fields and storyboards in Tableau requires some training.

9. Conclusion

Visualizing housing market trends through an interactive Tableau dashboard gives real estate stakeholders a clear, data-driven view of what drives property value. By examining features such as years since renovation, house age, and structural characteristics alongside sale price, this project demonstrates how visual analytics can simplify complex market data into actionable insights. While the current solution relies on periodic data imports and manual cleaning, it lays a strong foundation for more automated, real-time housing market intelligence in the future.

10. Future Scope

- Integration with live property-listing feeds for real-time price tracking.
- Predictive pricing models using machine learning.
- Expansion to cover rental markets and commercial properties.
- Geographic / map-based visualization of price trends by neighbourhood.
- Automated alerts for anomalous pricing or high-potential renovation opportunities.
- Mobile-first dashboard experience for on-the-go analysts.

11. Appendix

Tableau Dashboard link :

https://public.tableau.com/views/HousingMarketTrendsDashboard_17839591336410/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story Link: https://public.tableau.com/shared/Q2F4Z6P2Y?:display_count=n&:origin=viz_share_link

GitHub Link: <https://github.com/Ishwaryambika92/Visualizing-Housing-Market-Trends-An-Analysis-of-Sale-Prices-and-Features-using-Tableau>

Video Demo Link: https://drive.google.com/file/d/1876R_zSa4A-8zK48td_X85cLlqwquHUI/view?usp=drivesdk